On the Optimality of Progressive Income Redistribution

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Redistributive Taxation

- Why redistribution ?
  - Equity
  - Insurance

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  - Insurance
- Why focus on generations ?
  - 60 - 90% of wage inequality is explained by heterogeneity at the time of market entry. (Keane and Wolpin '97, Haider '01, Storesletten et al. '04)
  - Life-cycle shocks may be largely transitory (Guvenen '09).

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  - Life-cycle shocks may be largely transitory (Guvenen ‘09).
- Why public insurance ?
  - Private insurance protects against shocks that occur *after* a contract is signed.
  - Parents cannot accept obligations for their kids.

Question : How progressive should the tax system be ?

- Environment
  - Dynastic Setting
  - Intergenerational income risk
- Approach
  - Ramsey with commitment
  - Quantitative
- Constraints :
  - Public insurance affects private (self-)insurance
  - Incentives : *labor supply, savings*
  - Equilibrium effects : *interest rate, wage rate*

Literature

- **Life-cycle**
  - **Non-linear / Progressive Income Taxation :** Conesa et al. *(AER '09)*, Conesa and Krueger *(JME '08)*, Heathcote et al. ('10), Mirrlees *(RES '71)*, Saez (2001), Kocherlakota *(Econometrica '05)*, Albanesi and Sleet *(RES '06)*, Farhi and Werning *(JPE '12)*
  - **Private vs. Public Insurance :** Krueger and Perri *(JET '11)*, Hubbard et al. *(JPE '96)*, Attanasio and Rios-Rull *(EER '00)*, Cutler and Grueber *(QJE '96)*, Chetty and Saez (AEJ : EP '10), Golosov and Tsyvinski *(QJE '07)*
- **Dynastic :** Seshadri and Yuki *(JME '04)*, Benabou *(Econometrica '01)*
- Capital/Estate Taxation : ...
- Davila, Hong, Krusell and Rios-Rull *(Econometrica '12)*

Model

Model : Aiyagari '94 at Dynastic Frequency

– Consumer's Problem

$$U(k_t, z_t) = \max_{k_{t+1}, n_t, c_t} \mathbb{E} \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}}{1-\sigma} - \theta \frac{n_t^{1+\epsilon}}{1+\epsilon} \right]$$

subject to

$$\begin{array}{rcl} c_t + k_{t+1} & = & y^d + k_t \\ k_{t+1} & \geq & 0 \\ n_t & \in & [0,1] \\ z_{i,t+1} & \sim & F(z_{i,t+1} | z_t) \\ k_0, z_{i0} & are & given \end{array}$$

Redistributive Taxation

$$y^d = \lambda(w_t z_t n_t + r_t k_t)^{1-\tau}$$

- $\tau \leq 1$ measures the degree of progressivity of the tax system.
  - $\tau = 1:$ Full redistribution
  - $0 < \tau < 1:$ Partial redistribution
  - $\tau = 0:$ Flat tax
  - $\tau < 0:$ Regressive redistribution
- Marginal rate is monotonic in pre-tax income.
- Permits net transfers

  ... examples : Welfare-to-work (Workfare) and EITC

Model

– Production and Firms

$$\max_{K_t, N_t} \; K_t^{\alpha} N_t^{1-\alpha} - w_t N_t - (r_t + \delta) K_t$$

– Aggregates and Equilibrium

$$\begin{aligned} K_{t+1} &= \int k_{t+1} d\Gamma(k_t, z_t) \\ N_t &= \int z_t n_t d\Gamma(k_t, z_t) \end{aligned}$$

– Government's Budget

$$\int \left[y_t(k_t, z_t) - y^d(y_t(k_t, z_t)) \right] d\Gamma(k_t, z_t) = G_t = \gamma Y_t$$

Social Welfare

- Long-run optimum

$$\mathcal{W}^{ss} = \int V^{ss}(k,z;\Gamma^{ss})d\Gamma^{ss}(k,z)$$

- Optimal *once-and-for-all* tax reform starting at the status quo

$$\mathcal{W}(\Gamma) = \max_{\lambda,\tau} \int U(c(k,z;\Gamma),h(k,z;\Gamma))d\Gamma(k,z) + \beta\mathcal{W}(\Gamma')$$

subject to

$$\begin{aligned} \gamma Y &= \int [y - y^d(y;\lambda,\tau)]d\Gamma(k,z) \\ y &= wzh(k,z;\Gamma) + rk'(k,z;\Gamma) \\ \Gamma' &= H(\Gamma), \Gamma_0 \text{ given.} \end{aligned}$$

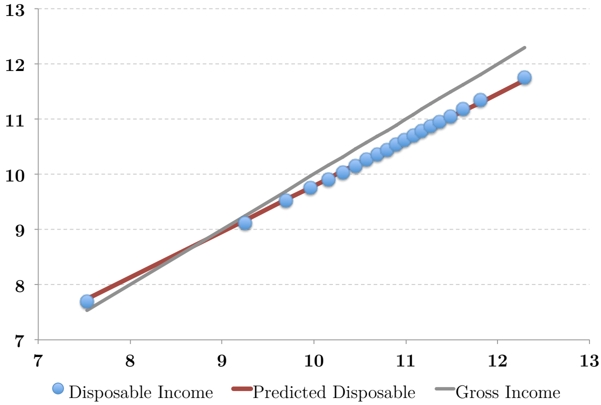
Calibration

Calibration

Parameter	*Value*	*Target Moment*	
T	$= 25$	Length of a generation	
σ	$= 2.00$	relative risk aversion	2.00
$\beta^{1/T}$	$= 0.965$	annual interest rate	4.3%
α	$= 0.360$	capital share of income	0.36
δ	$= 0.876$	annual depreciation rate	0.08
θ	$= 0.305$	average annual labor hours	0.44
ϵ	$= 0.825$	coef. of variation of hours	0.29
G/Y	$= 0.250$	authors' estimates	

Calibration : The US Tax System

- Data : March CPS (1977 - 2009) + NBER Tax Simulator
- Disposable Inc. = Gross Inc. - (Federal + State + Payroll Tax)



$$\log y^d = 1.34 + 0.83 \log y \qquad R^2 = 0.94 \qquad \tau = 0.17$$

Calibration : Intergenerational Wage Mobility in US

- Data : PSID, household heads, fathers and sons, 24-60 years of age.
- Wages : estimated fixed effects

Quartile	1	2	3	4
1	0.34	0.33	0.17	0.16
2	0.18	0.31	0.34	0.17
3	0.12	0.29	0.31	0.28
4	0.07	0.16	0.24	0.53
wage rate ($/hr)	9.7	14.9	20.2	33.0

- Hours : estimated fixed effects for men of ages 16 - 75.

Mean	1908 hrs/year	Coef. of variation	0.29
$\Rightarrow$	$\theta = 0.305$	$\epsilon = 0.825$	

Calibration : Implications

- Elasticity of Labor Supply

	Model		Literature
Uncompensated Wage Elasticity	0.42	*(0.10)*	0 to 0.3
Income Elasticity	-0.72	*(0.10)*	0 to -1
"Frisch" Elasticity	0.84		–

- Intergenerational correlation of hours
  - Model : 0.85 Data : 0.20 ▸ Transitions
- Intergenerational correlation of wealth
  - Model : 0.50 Data : 0.37 ▸ Transitions

Steady State Results and Analysis

Optimal Tax System : Steady State Comparison

	Benchmark (US)	Flat Tax	Optimal
Progressivity (τ)	0.17	0.00	-0.38
$\mathbb{E}[n]$	0.44	0.46	0.52
N	44.0	46.2	50.8
K	1.87	2.47	3.98
Y	14.1	16.1	20.3
$\mathbb{E}[c]$	9.4	10.5	12.6
r	4.27	3.68	2.74
w	0.21	0.22	0.26
$\mathcal{W}^{ss}$	-0.255	-0.245	-0.238

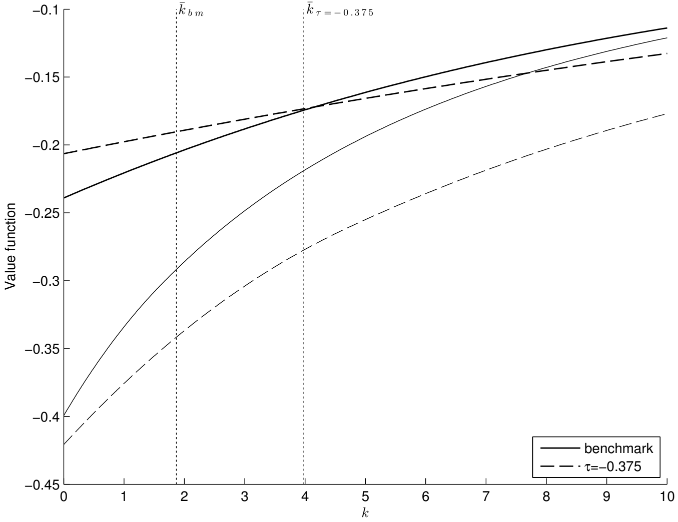
Optimal Tax System : Averages and Inequality relative to the Benchmark

	change in	
	average(%)	Gini
k	113.0	0.08
y	44.0	0.01
y^d	34.0	0.09
c	34.0	0.05
$1-n$	-11.1	0.04
$\mathcal{W}^{ss}$	9.1	0.06

Progressive Taxation

- A less progressive tax system ($\tau_L \searrow$)...
  - ... raises the return to working and saving
  - ... raises the *volatility* of idiosyncratic productivity shocks.
- Larger wealth and labor supply affect equilibrium prices and income.
- Bequests help absorb the additional risk.

Welfare by Wealth and Productivity



Three Counterfactual Experiments

- The role of precautionary saving
- The role of prices
- The role of labor supply

Counterfactuals : The Role of Precautionary Saving

- Keep the savings policy fixed at its benchmark value.
  - This prevents the savings response to tax policy.
  - Lower τ implies higher inequality without compensation from higher consumption.
  - Optimal τ : 0.27

Counterfactuals : The Role of Precautionary Saving

- Keep the savings policy fixed at its benchmark value.
  - This prevents the savings response to tax policy.
  - Lower τ implies higher inequality without compensation from higher consumption.
  - Optimal τ : 0.27
- No private capital : Government holds benchmark capital stock, rebates capital income to workers.
  - No self-insurance at all : hand-to-mouth consumers
  - Labor supply adjustment to tax policy
  - Optimal τ : 0.46

Counterfactuals : Equilibrium Effects and Labor Supply

- Fix r and w
  - Higher savings and labor supply do not generate a price response.
  - No built-in redistribution.
  - Optimal τ : 0.23

Counterfactuals : Equilibrium Effects and Labor Supply

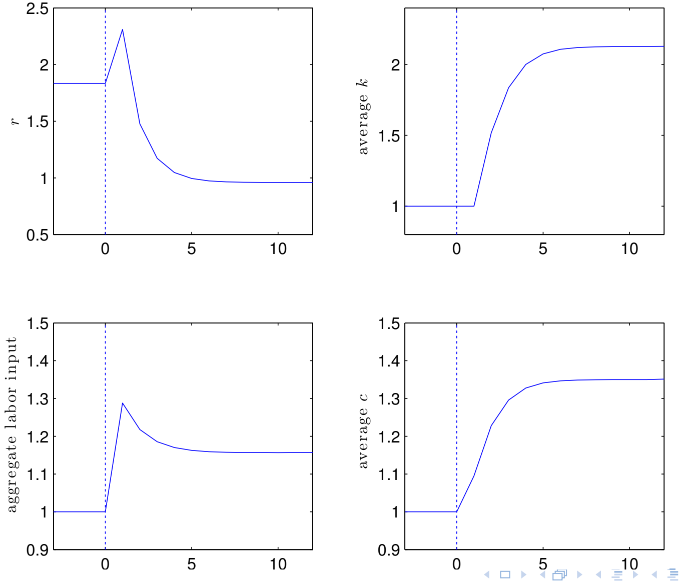
- Fix r and w
  - Higher savings and labor supply do not generate a price response.
  - No built-in redistribution.
  - Optimal τ : 0.23
- Also fix labor supply
  - Redistribution is less costly.
  - (Partial) savings adjustment to redistribution.
  - Optimal τ : 0.31

Transition Dynamics and Optimality

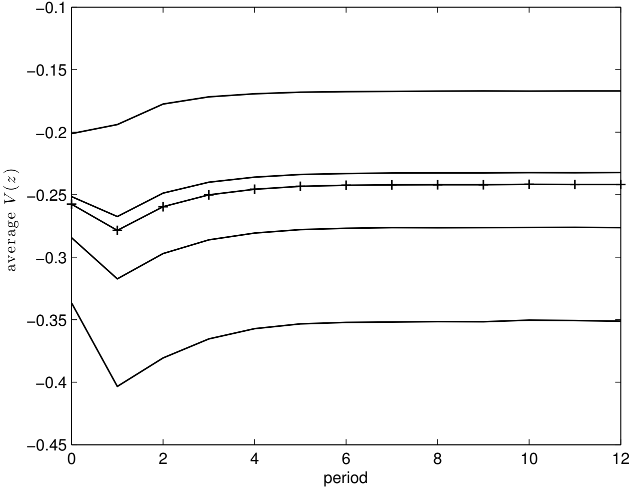
Optimal Progressivity and Transition Dynamics

- Starting at the current system, is it optimal to implement the steady-state policy ?
- If not, what is the optimal tax reform ?

Transition to Regressive Taxation



Value and Behavior along the Transition

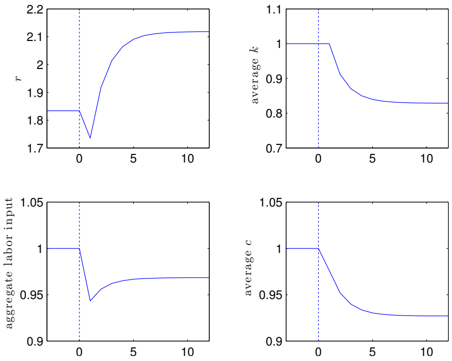


(a) Value

Political Support for the Regressive Tax Reform

- overall : 30%
- by productivity : <1%, 3%, 13%, 82%
- by wealth tertiles : <1%, 30%, 57%

Optimal Tax Reform : $\tau^* = 0.26$



+ immediate increase in leisure
+ immediate drop in income and consumption inequality
– lower long-run consumption.

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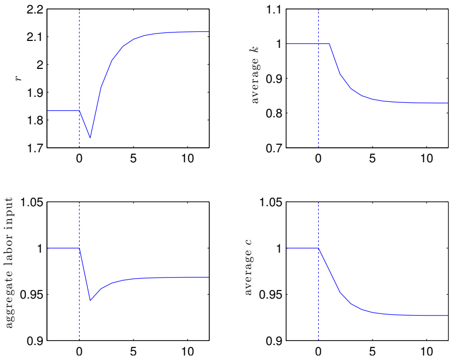

+ immediate increase in leisure
+ immediate drop in income and consumption inequality
– lower long-run consumption.

Political Support : 48%

- by productivity : 95%, 80%, 46%, 0%
- by wealth : 81%, 51%, 13%

Discussion

- Short-run vs. long-run optimality
- Anticipation and optimal tax policy
- Separate Capital Taxation ▸ Details
- Insurance vs. Egalitarianism ▸ Details
- Life-cycle income risk ▸ Details

APPENDIX

Separate capital taxation

Consider $y^d = \lambda(wzh)^{1-\tau_l} + (1-\tau_k)rk$:

	τ_l	τ_k
optimal :	-0.19	-1.125
no K taxes, optimal τ_l :	-0.23	0
flat L taxes, optimal τ_k :	0	-1.0

- Optimal labor taxes remain regressive with separate K taxes/subsidy.
- Regressive labor taxes complement a K subsidy.

Life-Cycle Income Risk and Optimal Progressivity

- time period : 5 years
- death probability : 20%
- transitions : dynastic if dead, life-cycle otherwise
  - estimated from PSID
  - more persistent per period, less persistent overall in 25 years.
- Optimal τ : -0.09
  - less persistence - less benefit from redistribution
  - less precautionary savings - smaller savings response.

Social Welfare Criteria : Insurance vs. Egalitarianism

- Certainty Equivalence Criteria (Benabou '02)

$$\begin{aligned} V_0(k,z) &= \sum_{t=0}^{\infty} \beta^t U(\tilde{c}(k,z), H_t) \\ \tilde{C}_t &= \int \tilde{c}(k,z) d\Gamma_t(k,z) \\ \mathcal{W}^{\mathcal{P}} &= \sum_{t=0}^{\infty} \beta^t U(\tilde{C}_t, H_t) \end{aligned}$$

- Optimal tax reform is still regressive.

Intergenerational Wealth Transitions

Model

	1	2	3	4	5
1	0.57	0.26	0.17	0.00	0.00
2	0.43	0.33	0.11	0.13	0.00
3	0.07	0.28	0.34	0.31	0.00
4	0.00	0.04	0.38	0.26	0.33
5	0.00	0.00	0.02	0.31	0.67

Charles and Hurst 2002

	1	2	3	4	5
1	0.36	0.26	0.16	0.15	0.11
2	0.29	0.24	0.21	0.13	0.16
3	0.16	0.24	0.25	0.20	0.14
4	0.12	0.15	0.24	0.26	0.24
5	0.07	0.12	0.15	0.26	0.36

Intergenerational Hours Transitions

Model

	1	2	3	4
1	0.87	0.13	0.00	0.00
2	0.13	0.70	0.15	0.02
3	0.00	0.21	0.69	0.09
4	0.00	0.00	0.14	0.86

PSID

	1	2	3	4
1	0.28	0.25	0.26	0.22
2	0.22	0.25	0.31	0.23
3	0.18	0.26	0.28	0.28
4	0.15	0.20	0.27	0.38